

Description & Eligibility

Demonstrate knowledge of fundamental database development and SQL scripting. Competencies addressed in this event will mandate the contestant use a high-end database product such as MS SQL Server®. The focus of this event is to acquire the necessary skills; however, topics addressed are transferable to any database product such as Oracle® or MySQL™. Any postsecondary or secondary division contestant may enter this objective event.

Contestant Must Supply

- Published and/or unpublished non-electronic written reference materials
- Cordless calculator: Electronic devices will be monitored according to ACT standards. See NLC Calculator Guidelines. Contestants who violate this rule will be *disqualified*.

Competition Notes

- Method of evaluation: objective.
- Previous Business Professionals of America tests and/or sample tests (handwritten, photocopied or keyed) may not be taken into the testing area. **Violation of this rule will result in disqualification.**
- At NLC, contestants will take IT Specialist - Databases certification as part of the event. Contestants passing this certification will have 50 points added to their event score. A Certiport profile must be created at www.certiport.com before attending NLC. Contestants must include their BPA Contestant ID in their Certiport profile when they register. For more information on the exam, visit www.certiport.com. Reference materials are not allowed for certification testing.
- Length of event: no more than ten (10) minutes orientation, no more than sixty (60) minutes testing time, and no more than ten (10) minutes wrap-up.
 - No more than one hundred twenty (120) minutes for certification test.
- Entries vary by state.

Contest Competencies

- Identify various types of databases
- Create and employ tables, rows, columns
- Understand roll of keys, constraints
- Understand and create indexes
- Define relational types and integrity
- Identify normal forms and normalize to 3NF
- Understand data types and when to use them
- Understand SQL Server® schemas
- Understand/use DDL commands such as CREATE, DROP, ALTER database
- Understand/use DML commands such as INSERT, SELECT, UPDATE, DELETE
- Select data from multiple tables using the WHERE clause
- Understand JOIN, UNION, INTERSECT
- Use aggregate and scalar functions
- Create and use views
- Use and understand transactions
- Understand how to back up databases
- Understand Stored Procedures